

Introduction to Data Science

Passing the subject:

- hand-in assignments
- final exam in January
 - computer based
 - 20/40 points
 - four additional programming assignments, each 24 points, you can get up to 6 points on the main exam for solving them

Lecture 1

Definition: An experiment is an activity that produces a distinct outcome.

- Ω - sample space that contains all possible outcomes

Ex: Tossing a coin

$$\Omega = H, T$$

Event: a set of outcomes

An event happens after an experiment if the outcome $\omega \in \Omega$; $\omega \in A$

What is probability?

(Idea) (Long term relative frequency) Let's repeat an experiment n -times and count how many times an event A happens

$$N(A, n) = \frac{\text{number of times } A \text{ happens}}{n}$$

- assume the experiment is independent

Example: if $N(A, n) \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$ then we could say "50%" chance of happening probability

Rules

1. $N(\Omega, n) = 1$ - something happens
2. Take two events, A, B . $A \cap B = \emptyset$, then $N(A \cup B, n) = N(A, n) + N(B, n)$

Definition: A function $\mathbb{P}$ that assigns real values to each event is a probability "distribution"/"measure"

if $0 \leq \mathbb{P} \leq 1$

1. $\mathbb{P}(\Omega) = 1$
2. A, B $A \cap B = \emptyset$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$

some other notations and cases:

- A and A' (complement) $\mathbb{P}(A) + \mathbb{P}(A') = 1$
- A and B intersect $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)$

Bolles inequality (union bound):

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$$

Conditional probability

What is the probability of A , given that I know B happens?

- B becomes the new Ω

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Example: SMS messages

- A - "message is spam"
- B - "free appears in the text"

Law of total probability

Let $A_1, \dots, A_n$, $A_i \cap A_j = \emptyset$

$$\bigcup_{i=1}^n A_i = \Omega$$

then for any $B \in \Omega$

$$\mathbb{P}(B) = \sum_{i=1}^n \mathbb{P}(B|A_i)\mathbb{P}(A_i) = \sum_{i=1}^n \mathbb{P}(B \cap A_i)$$

basically to get probability of B, sum its intersections with A (A makes up the whole omega space)

Lecture 2

Random variables

A random variable (RV) is a mapping

$$X : \Omega \rightarrow \mathbb{R}$$

that assigns a real number $X(\omega)$ to each outcome ω .

$$\mathbb{P}(X \leq 1) := \mathbb{P}(\omega \in \Omega : X(\omega) \leq 1)$$

all outcomes such that $X(\omega) \leq 1$

Example.

$\Omega = \{\text{all possible texts of length 100}\}$

Experiment: Receiving a text of length ≤ 100 ω

$X(u) = 1$ if omega is spam, 0 if not

$$X(u) = \begin{cases} 1 & \text{if } \omega \text{ is spam} \\ 0 & \text{if not} \end{cases}$$

$$P(X = 1) = P(\omega \text{ is spam})$$

Cumulative distribution function CDF

$$F_X : \mathbb{R} \rightarrow [0, 1] \quad F_X(x) = P(X \leq x)$$

Properties of CDF:

- non decreasing $x_1 < x_2$ implies that $F_X(x_1) \leq F_X(x_2)$
- F is normalized $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$
- F is right continuous
- CDF gives the probability that an RV X will take a value less than or equal to a specific value x

If X is discrete, we can define a **probability mass function** (PMF)

$$f_X(x) = P(X = x)$$

Connection to the CDF:

$$F_X(x) = \sum_{t \leq x} f_X(t)$$

For a continuous random variable we distinguish a **probability density function** (PDF) $f_X(x)$. It holds that for PDF:

$$F_X(x) = \int_{-\infty}^x f_X(t)dt \text{ and } f_X(x) = F'_X(x)$$

- PDF describes the relative likelihood for a continuous random variable to take on a specific value
- this is not a probability, it can be greater than 1
- total area under the curve must be 1

Inverse CDF (quantile function)

$$F^{-1}(q) = \inf x : F(x) > q \quad q \in [0, 1]$$

If F is strictly increasing and continuous then $F^{-1}(F(x)) = x$ and $F(F^{-1}(q)) = q$

- Inverse CDF takes probability q and returns the corresponding value of x (an outcome)

We call

- first quartile $F^{-1}(0.25)$
- median/second quartile $F^{-1}(0.5)$
- third quartile $F^{-1}(0.75)$

Transformations of random variables

X is an RV with PDF f_X and CDF F_X . Let $Y = g(X)$ be a transformation of X. How do we compute PDF and CDF of Y?

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right|$$

Some important discrete random variables

Point Mass Distribution

$$X \sim \delta_a \text{ if } P(X = a) = 1$$

$$F(X) = \begin{cases} 0 & x < a \\ 1 & x \geq a \end{cases} \quad f(x) = \begin{cases} 1 & \text{if } x = a \\ 0 & \text{else} \end{cases}$$

Discrete Uniform distribution

$$f(x) = \begin{cases} \frac{1}{k} & \text{for } x = 1, \dots, k \\ 0 & \text{otherwise} \end{cases}$$

Bernoulli distribution $P(X=1) = p, P(X=0) = 1-p$

$$X \sim \text{Bernoulli}(p) \quad f(x) = p^x (1-p)^{1-x}$$

Binomial distribution Number of successes in n independent Bernoulli trials with success probability p

$$X \sim \text{Binomial}(n, p) \quad f(x) = \binom{n}{x} p^x (1-p)^{n-x} \text{ for } x = 0, 1, \dots, n$$

x - number of successes

Geometric Distribution

$$X \sim \text{Geom}(p) \quad P(X = k) = p(1-p)^{k-1}, \quad k \geq 1$$

- interpretation: number of trials until first success in independent Bernoulli trials with success probability p

Poisson Distribution

$$X \sim \text{Poisson}(\lambda) \quad f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \geq 0$$

Some important continuous random variables

Uniform Distribution

$$X \sim U(a, b) \quad f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Distribution function is

$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

Normal Gaussian Distribution

$$X \sim N(\mu, \sigma^2) \quad f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp -\frac{1}{2\sigma^2}(x - \mu)^2$$

- standard normal has sigma=1 and mean=0

$$X \sim N(\mu, \sigma^2) \text{ then } Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

- sum of gaussian independent RVs has the following distribution

$$\sum X_i \sim N(\sum \mu_i, \sum \sigma_i^2)$$

We can compute any probabilities as long as we compute the CDF $\Phi(z)$ of standard Normal

Exponential distribution

$$X \sim \text{Exp}(\beta) \quad f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x > 0$$

Lecture 3

Expectation

Toss a dice 100 times

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$$\frac{1 + 3 + 5 + 2 + \dots + 3}{100} = 3.1 \text{ (?) } \$$$

$$\frac{1}{100} * 1 + \frac{2}{100} * 2 + \dots + \frac{6}{100} * 6 = 1 * P(X=1) + 2 * P(X=2) + \dots + 6 * P(X=6)$$

$$E[X] = \sum x \cdot f_X(x) \quad \text{expectation} \quad \text{remainder} \quad f_X(x) - P(X=x) \quad \$$$

What about random vectors? X is R^m and discrete

$$E[X] = \sum x \cdot f_X(x) = \sum_{x_1, x_2, \dots, x_m} x_1 f_X(x_1, x_2, \dots, x_m), \dots, \sum_{x_1, x_2, \dots, x_m} x_m f_X(x_1, x_2, \dots, x_m) \quad \$$$

Linearity of expectation

Say we have X and Y RV's $Z = (X, Y)$

$$E[Z] = E[X + Y] = \sum_{x,y} (x+y) f_{X,Y}(x,y) = \sum_{x,y} x f_{X,Y}(x,y) + \sum_{x,y} y f_{X,Y}(x,y) = E[X] + E[Y] \quad \$$$

Law of the lazy statistician (LOLS)

Let's say we have a random variable X and a function $f: \mathbb{R} \rightarrow \mathbb{R}$

Let $Y = f(X)$

$$E[Y] = \sum_y y f_Y(y) = E[f(X)] = \sum_x f(x) f_X(x)$$

$f_Y(y)$ might be harder to find than $f_X(x)$

Variance

$E[(X - E[X])^2] = E[X^2] - (E[X])^2$ variance is the expectation of the squared deviation from the mean

Mean absolute deviation

$$\mathbb{E}[|X - \mathbb{E}[X]|]$$

If $X_1, X_2, \dots, X_n$ are independent then the variance of the sum is the sum of the variances

Covariance

$$Cov(X, Y) = \mathbb{E}((X - \mathbb{E}[X])(Y - \mathbb{E}[Y])) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

how related are X and Y, how strong the linear relationship is Correlation is normalized covariance $\rho_{X,Y} = \frac{Cov(X,Y)}{\sqrt{V(X)V(Y)}} \in [-1, 1]$

If X and Y are RVs then

$$V(X + Y) = V(X) + V(Y) + 2Cov(X, Y)$$

If X and Y are independent then

$$V(X + Y) = V(X) + V(Y)$$

Mean and variance of some important distributions

Distribution	Mean	Variance
Point mass at a	a	0
Bernoulli(p)	p	$p(1 - p)$
Binomial(n, p)	np	$np(1 - p)$
Geometric(p)	$1/p$	$(1 - p)/p^2$
Poisson(λ)	λ	λ
Uniform(a, b)	$(a + b)/2$	$(b - a)^2/12$
Normal(μ, σ^2)	μ	σ^2
Exponential(β)	β	β^2
Gamma(α, β)	$\alpha\beta$	$\alpha\beta^2$
Beta(α, β)	$\alpha/(\alpha + \beta)$	$\alpha\beta/((\alpha + \beta)^2(\alpha + \beta + 1))$
t_ν	0 (if $\nu > 1$)	$\nu/(\nu - 2)$ (if $\nu > 2$)
χ_p^2	p	$2p$
Multinomial(n, p)	np	skibidi
Multivariate Normal(μ, Σ)	μ	Σ

Tower rule

$$\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$$

Cauchy distribution

$$f_X(x) = \frac{1}{\pi(1 + x^2)}$$

It's mean does not exist because Cauchy has thic tails and hence extreme observations are common

Moments of random variables

The k-th moment of a random variable X is defined as

$$\mu'_k = \mathbb{E}[X^k]$$

The k-th central moment of a random variable X is defined as

$$\mu_k = \mathbb{E}[(X - \mathbb{E}[X])^k]$$

Moment generating function (MGF)

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_x e^{tx} f_X(x)$$

We obtain the moments of X by differentiating MGF and evaluating at t=0

$$\mu'_k = M_X^{(k)}(0)$$

Notable moments

- 1st moment = expectation
- 2nd central moment = variance
- 3rd central moment = skewness - asymmetry of the distribution
- 4th central moment = kurtosis - "tailedness" of the distribution, measure of the heaviness of the tails, e.g.
 - normal distribution kurtosis = 3
 - Cauchy distribution kurtosis = infinity
 - exponential distribution kurtosis = 9
 - the higher the kurtosis, the heavier the tails -> more prone to outliers and slower convergence to the mean

If kth moment exists and if j < k then the jth moment also exists

If $X_1, \dots, X_n$ are random variables then we define the sample mean and sample variance as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

If $X_1, \dots, X_n$ are iid with mean μ and variance σ^2 then

$$\mathbb{E}[\bar{X}_n] = \mu \quad \mathbb{V}[\bar{X}_n] = \frac{\sigma^2}{n} \quad \mathbb{E}[S_n^2] = \sigma^2$$

Lecture 4

Markov inequality

X - non negative RV,

$$P(X \geq \epsilon) \leq \frac{\mathbb{E}[X]}{\epsilon} \quad \epsilon > 0$$

Chebyshev inequality

$$P(|X - \mathbb{E}[X]| \geq \epsilon) \leq \frac{\mathbb{V}[X]}{\epsilon^2}$$

- requires finite variance
- polynomial decay

Hoeffding inequality

Let $X_1, X_2, \dots, X_n$ be independent RVs such that $a_i \leq X_i \leq b_i$ almost surely. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then for any $\epsilon > 0$ it holds that

$$P(|\bar{X}_n - \mathbb{E}[\bar{X}_n]| \geq \epsilon) \leq 2 \exp\left(-\frac{2n\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

confidence intervals:

Set the RHS equal to δ and solve for ϵ :

- Hoeffding:

$$2e^{-2n\epsilon^2/(b-a)^2} = \delta \Rightarrow \epsilon = \frac{b-a}{\sqrt{2n}} \sqrt{\ln\left(\frac{2}{\delta}\right)}.$$

- Chebyshev:

$$\frac{\sigma^2}{n\varepsilon^2} = \delta \Rightarrow \varepsilon = \sqrt{\frac{\sigma^2}{n\delta}}.$$

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Sub-Gaussian and Sub-Exponential random variables

Let X be RV. We call it sub-Gaussian with parameter λ if

$$E[e^{t(X-E[X])}] \leq e^{\frac{\lambda^2 t^2}{2}} \text{ for all } t \in \mathbb{R}$$

We call it sub-Exponential with parameters (ν, b) if

$$E[e^{t(X-E[X])}] \leq e^{\frac{\nu^2 t^2}{2}} \text{ for } |t| < \frac{1}{b}$$

- here we bound the moment generating function
- SG RVs have tails that decay at least as fast as Gaussian tails
- SE tail decays slower than Gaussian but still exponentially
- every SG is SE but not vice versa
- if X is SG then X^2 is SE
- bounded RVs are sub-Gaussian
- sum of SG RVs is SG

Types of convergence of random variables

- convergence in probability

$$X_n \xrightarrow{P} X \text{ if for any } \epsilon > 0 \quad P(|X_n - X| \geq \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

- convergence almost surely

$$X_n \xrightarrow{a.s.} X \text{ if } P(\lim_{n \rightarrow \infty} X_n = X) = 1$$

- strongest form of convergence
- X_n converges to X for almost every outcome ω in the sample space
- convergence in distribution

$$X_n \xrightarrow{d} X \text{ if } F_{X_n}(x) \rightarrow F_X(x) \text{ for all points } x \text{ where } F_X \text{ is continuous}$$

- sequence of CDFs approaches the limit's CDF (the shape becomes the same)
- convergence in L_p

$$\lim_{n \rightarrow \infty} E[|X_n - X|^p] = 0$$

Hierarchy of implications:

- almost surely $\Rightarrow$ probability $\Rightarrow$ distribution
- $L_p \Rightarrow$ probability

Weak law of large numbers (WLLN)

Let $X_1, X_2, \dots, X_n$ be iid RVs with mean μ and variance $\sigma^2 < \infty$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then for any $\epsilon > 0$ it holds that

$$\bar{X}_n \xrightarrow{P} \mu \text{ as } n \rightarrow \infty \iff \lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \epsilon) = 0$$

- requires only finite variance

Strong law of large numbers

Let $X_1, X_2, \dots, X_n$ be iid RVs with mean μ . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then it holds that $\bar{X}_n \xrightarrow{a.s.} \mu$ as $n \rightarrow \infty$

- requires finite mean (first moment)

Central limit theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be iid RVs with mean μ and variance $\sigma^2 < \infty$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then it holds that $Z_n = \frac{\bar{X}_n - E[\bar{X}_n]}{\sqrt{\text{Var}(\bar{X}_n)}} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$

Lesson 2

Lecture 5 - Estimation

Bias

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

MSE

$$\text{MSE} = \mathbb{E}[(\hat{\theta} - \theta)^2] = V[\hat{\theta}] + \text{Bias}(\hat{\theta})^2$$

DKW Inequality

Let $X_1, X_2, \dots, X_n$ be iid RVs with CDF F . Let F_n be the empirical CDF. Then for any $\epsilon > 0$ it holds that

$$P(\sup_x |F_n(x) - F(x)| > \epsilon) \leq 2e^{-2n\epsilon^2}$$

A non-parametric $1-\alpha$ confidence band for F

$$L(x) = \hat{F}(x) - \epsilon, U(x) = \hat{F}(x) + \epsilon \text{ where } \epsilon = \sqrt{\frac{1}{2n} \ln\left(\frac{2}{\alpha}\right)}$$

Lecture 6 - Random variable generation

Congruential generator

Parameters:

- a - multiplier
- b - increment
- m - modulus

$$D(x) = (ax + b) \mod m$$

Hull-Dobell theorem

Congruential generator (a, b, M) has period M iff:

- $\gcd(b, M) = 1$
- p divides $a-1$ for every prime p that divides M
- 4 divides $a-1$ if 4 divides M

Examples of (a, b, M) :

- $(1103515245, 12345, 2^{32})$
- $(29, 12345, 2^{32})$

Accept-Reject sampler

To sample from a distribution with PDF $f(x)$:

1. Choose a proposal distribution with PDF $g(x)$ such that $f(x) \leq c \cdot g(x)$ for some $c > 0$
2. Repeat:
3. Sample Y from $g(x)$
4. Sample U from $\text{Uniform}(0,1)$
5. If $U \leq f(Y)/(c \cdot g(Y))$, return Y ; else, go back to step 3
6. The accepted samples follow the distribution with PDF $f(x)$

Bax-Muller

To generate standard normal RVs:

1. Generate $U_1, U_2 \sim \text{Uniform}(0,1)$
2. Compute:

$$Z_0 = \sqrt{-2 \ln(U_1)} \cos(2\pi U_2) \quad Z_1 = \sqrt{-2 \ln(U_1)} \sin(2\pi U_2)$$

3. Z_0 and Z_1 are independent standard normal RVs

Inverse transform sampling

Goal: Generate random samples from a target distribution X using a source of uniform random numbers $U \sim \text{Unif}(0,1)$.

Prerequisites:

1. You must know the Cumulative Distribution Function (CDF), $F(x)$.
2. You must be able to calculate the inverse function, $F^{-1}(u)$.

The Algorithm

1. **Generate** a random number u from the uniform distribution $[0,1]$.
2. **Compute** $x = F^{-1}(u)$.
3. **Result:** The value x is a random draw from the distribution defined by F .

Mathematical Proof

Let $U \sim \text{Unif}(0,1)$. We define $X = F^{-1}(U)$. We verify the CDF of X :

$$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) \quad (\text{Applying } F \text{ to both sides}) = F(x) \quad (\text{Property of Uniform distribution})$$

Conclusion: The variable X has the CDF $F(x)$, meaning it follows the desired distribution.

Example: Exponential Distribution

- **PDF:** $f(x) = \lambda e^{-\lambda x}$
- **CDF:** $F(x) = 1 - e^{-\lambda x}$
- **Inverse Step:** Let $u = 1 - e^{-\lambda x}$

$$e^{-\lambda x} = 1 - u$$

$$-\lambda x = \ln(1 - u)$$

$$x = -\frac{1}{\lambda} \ln(1 - u)$$

- **Sampling Formula:** Generate u , then compute $x = -\frac{\ln(1-u)}{\lambda}$.

Other notes

Train-Validation-Test splitting

IF we want to achieve 60/15/25 split:

1. First get 40% of the data as a 'temp' set and 60% as train set
2. Then split the 'temp' set into validation and test sets according to the formula

$$test_size = \frac{P_{test}}{P_{test} + P_{val}}$$

Bennett's inequality

Let $X_1, X_2, \dots, X_n$ be independent, centered (mean 0) sub-Exponential RVs with parameters (ν_i, b_i) . Let $S_n = \sum X_i$. Then for any $t > 0$ it holds that

$$P(S_n \geq t) \leq \exp\left(-\frac{n\nu_n^2}{b_n^2} h\left(\frac{b_n t}{n\nu_n^2}\right)\right)$$

where

$$\nu_n^2 = \frac{1}{n} \sum_{i=1}^n \nu_i^2, \quad b_n = \max_i b_i, \quad h(u) = (1+u) \ln(1+u) - u$$

Bernstein's inequality is a simpler version of Bennett's inequality

$$P(S_n \geq t) \leq \exp\left(-\frac{t^2/2}{n\nu_n^2 + b_n t/3}\right)$$

Epsilon confidence interval:

$$\varepsilon = \sqrt{\frac{2\sigma^2}{n} \ln\left(\frac{2}{\delta}\right)} + \frac{M}{3n} \ln\left(\frac{2}{\delta}\right)$$

σ^2 - average of variances of X_i M = maximum possible range/bound of X_i δ - confidence level (e.g. 0.01)

Additional notes at: <https://bluej1.github.io/IntroDSEexamMaterial/>

solutions to exercises: https://github.com/sajad13901/Statistics_Wasserman/tree/main